

Octal -- Guided Notes ANSWER KEY

Unit tp-1.4 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

For instructor use / distribute after students complete the notes.

Question 1

Octal is a base-_____ number system. It uses only the digits _____ through _____. The digits 8 and 9 do _____ (do/do not) exist in octal. Each octal digit represents exactly _____ binary bits.

Answer: 8; 0 through 7; do not; 3.

Question 2

Convert the octal number 37 to decimal:

3 is in the 8^1 position = $3 \times$ _____ = _____

7 is in the 8^0 position = $7 \times$ _____ = _____

Total = _____

Answer: $3 \times 8 = 24$; $7 \times 1 = 7$; Total = 31.

Question 3

To convert binary to octal, group binary digits into sets of _____ starting from the right. Convert 110101:

Left group: _____ = octal _____

Right group: _____ = octal _____

Result: _____

Answer: 3 bits. Left: 110 = 6. Right: 101 = 5. Result: 65 (octal).

Question 4

In Linux, file permissions are displayed and set using octal. The permission value 755 means: owner has _____ (read/write/execute), group has _____ (read/execute only), and others have _____ (read/execute only). The octal digit 7 = binary _____ = rwx.

Answer: Read/write/execute (7); read/execute (5); read/execute (5); 111.

Question 5

Using octal, the chmod command sets file permissions on Linux. chmod _____ grants full permissions to owner, and read/execute only to group and others. chmod 600 means owner has _____ only, and group and others have _____ permissions.

Answer: 755; read/write (6 = rw-); no permissions (0 = ---).

Question 6

Compare the four number systems by filling in the equivalent values for decimal 10:

Binary: _____

Octal: _____

Decimal: 10

Hexadecimal: _____

Answer: Binary: 1010; Octal: 12; Decimal: 10; Hexadecimal: A.

Question 7

Octal was historically used in older computing systems because early computers used _____ -bit or _____ -bit groupings where octal mapped cleanly. Today, octal is less common than hex because modern systems are byte-oriented (8-bit), and 8 bits divides more cleanly into _____ hex digits than _____ octal digits.

Answer: 6-bit or 12-bit; 2 hex; 3 octal (8 bits = 2.67 octal digits -- not a clean boundary).

Question 8 -- Real World Application

REAL WORLD: A student is setting permissions on a web server script in Linux and runs `chmod 777` on the file. A senior admin immediately tells them to change it. Explain what `chmod 777` means using octal-to-binary conversion, why it is a security risk, and what a safer permission value would be and why.

Answer: chmod 777 means owner, group, and others all get binary 111 = read, write, and execute permissions. This is a security risk because any user on the system -- or any process running as any user -- can overwrite, delete, or execute the file. For a web server script, external users interacting with the web server process should generally only be able to read and execute (not write) the script. A safer value would be 755: owner gets rwx (7), group and others get r-x (5) -- they can read and run the script but cannot modify it.